

What Did Each Simulation Estimate?

AP Statistics · Unit 2 · Topic 2.3 · TEACHER KEY

CED TETHER

- **Skill 3.A** — Determine relative frequencies, proportions, or probabilities using simulation or calculations.
- **EU UNC-2** — Simulation allows us to anticipate patterns in data.
- **LO UNC-2.A** — Estimate probabilities using simulation. [Skill 3.A]
- **EK UNC-2.A.1** — A random process generates results that are determined by chance.
- **EK UNC-2.A.2** — An outcome is the result of a trial of a random process.
- **EK UNC-2.A.3** — An event is a collection of outcomes.
- **EK UNC-2.A.4** — Simulation is a way to model random events, such that simulated outcomes closely match real-world outcomes. All possible outcomes are associated with a value to be determined by chance. Record the counts of simulated outcomes and the count total.
- **EK UNC-2.A.5** — The relative frequency of an outcome or event in simulated or empirical data can be used to estimate the probability of that outcome or event.
- **EK UNC-2.A.6** — The law of large numbers states that simulated (empirical) probabilities tend to get closer to the true probability as the number of trials increases.

Essential question: How must chance labels, one trial, and the success event align with the probability question?

DOK-3 skill: 4.B · **FRQ pattern:** whose-simulation-design-is-valid-and-what-it-misestimates

DOK rationale: Part (c) weighs competing claims using the day's numerical or graphical evidence and explains a limitation or consequence; the judgment requires linked reasoning beyond a calculation.

Self-paced bonus sheet. Students take it when they choose and turn it in when they finish. Everything they need is printed on the sheet. Score part (c) only, E / P / I, by hand — never AI-graded or auto-scored. (a) and (b) are the ladder up; use them to see where a thin (c) came from.

Questions to ask

- Which number or visible feature supports your claim?
- Why does the other claim go beyond that evidence?
- What would you tell someone who chose the other claim?

[WARN] LOOK FOR

- A choice with copied numbers but no explanation of how they support it.
- Treating a model or average as a guaranteed individual outcome.

SCORING PART (c) — E / P / I

Expected elements

- **reasoning-1** (required) — Identifies the 0.10 versus 0.12 assignment mismatch
- **reasoning-2** (required) — Identifies exactly 2 versus at least 2 and rejects choosing by the larger estimate

- **reasoning-3** (required) — Gives five independent 00–99 pairs, 12 left-handed labels, and the at-least-2 event

E	Includes all three required reasoning elements above, with correct contextual evidence.
P	Includes at least one substantive correct reasoning link above, but omits or misstates another required element.
I	Includes no substantive correct reasoning link above; an unsupported choice or copied number alone is insufficient.

Common mistakes

- Counting digit 0 as 12 percent because 0 is one of the possible outcomes
- Treating exactly 2 and at least 2 as the same event
- Choosing Student 2 solely because two-digit numbers look more precise
- Fixing the label assignment but continuing to count exactly 2

[STAR] THE PROBLEM — annotated

Suppose each of five students can be treated as independently selected, and each has probability 0.12 of being left-handed. Two students use 1,000 simulation trials to estimate the probability that at least 2 of the 5 students are left-handed. Their plans and results are shown.

Results: Student 1 = 87/1,000; Student 2 = 94/1,000

Plan	L assignment	Trial; success
1	0 of 0–9	5 digits; 2+ L
2	00–11 of 00–99	5 pairs; exactly 2 L

FIRST TAKE — one sentence before you work the parts

Which simulation result would you trust more for the question asked—Student 1, Student 2, or neither? Name one reason.

Key: Not graded. Accept a tentative choice; the lesson should move students from decimal comparison to model-event alignment.

[BOOK] THREE MATCHES MAKE A VALID SIMULATION (keep on the page)

Match each outcome's chance with the fraction of equally likely labels assigned to it. One trial must reproduce the stated number of independent selections. Count the exact event named in the question. More trials reduce chance variation but do not repair a wrong assignment or event. Estimate probability with successful trials/total trials.

DOK 1 (a) List the left-handed counts included in at least 2 of 5. State the chance one student is left-handed.

Key: The event includes 2, 3, 4, or 5 left-handed students. A valid random-label assignment must designate 12 percent of equally likely labels as left-handed.

DOK 2 (b) Find each reported estimate. For each plan, identify either a mismatch in the chance assignment or a mismatch in the event counted.

Key: Student 1 reports $87/1000 = 0.087$ and Student 2 reports $94/1000 = 0.094$. Student 1 assigns only $1/10 = 0.10$ of the digits to left-handed, although the success event is correctly at least 2. Student 2 assigns $12/100 = 0.12$ of the pairs to left-handed, but counts exactly 2 rather than 2, 3, 4, or 5.

DOK 3 (c) Would you trust either plan for the question asked? Explain both mismatches and why the larger reported estimate does not validate a plan. Describe one corrected trial, including the random labels and the event counted. Write 3–4 sentences.

Key: Neither plan matches the question. Student 1 assigns $1/10 = 0.10$, not 0.12, to left-handed; Student 2 counts exactly 2 and leaves out 3, 4, and 5. A larger reported estimate does not fix either mismatch, and estimates also vary from run to run. A corrected trial draws five independent pairs from 00–99, uses 00–11 for left-handed and 12–99 for right-handed, and counts the trial when at least two selections are left-handed; repeat many trials and divide successes by trials.